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ThermoBreastAI: A Hybrid AI Library for Breast Thermography Classification

Rian Rabelo de Azevedo [1], **Adrielly Sayonara de Oliveira Silva** [2], **José Barbosa de Araújo Neto** [2], **Camila Tiodista de Lima** [2], **Flávio Secco Fonseca** [1], **Ana Clara Gomes da Silva** [2], **Clarisse Lima** [2], **Maíra Araújo de Santana** [2], **Juliana Carneiro Gomes** [2], **Giselle Machado Magalhães Moreno** [2], **and Wellington Pinheiro dos Santos** [1,2]

1 Graduate Program in Computer Engineering, University of Pernambuco, Brazil **2** Department of Biomedical Engineering, Federal University of Pernambuco, Brazil

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Software

- Review
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Summary

ThermoBreastAI is a Python library for computer-aided diagnosis of breast cancer using infrared thermography. It implements a hybrid artificial intelligence approach that combines transfer learning — using frozen pre-trained convolutional neural networks (MobileNetV2, VGG19, ResNet) for feature extraction — with classical machine learning classifiers (Random Forest, KNN, Naive Bayes) for diagnostic prediction. Originally developed to support the research presented in (Azevedo, 2025), (Santana et al., 2018), (Macedo et al., 2021), this library provides a modular, extensible, and reproducible framework for researchers working on thermal breast cancer screening, particularly valuable in low-resource settings where access to mammography is limited.

While general-purpose medical imaging libraries exist, few offer specialized, validated pipelines for thermographic analysis. **ThermoBreastAI** fills this gap by providing exact implementations of a hybrid AI workflow optimized for small medical datasets, enabling reproducible research and clinical decision support.

Statement of Need

Breast cancer is the most prevalent cancer among women worldwide (Sung et al., 2021). Despite advances in screening, challenges such as accessibility, patient discomfort, and radiation exposure remain — especially in developing regions (Instituto Nacional de Câncer, 2022). Infrared thermography offers a safe, non-invasive, low-cost alternative but suffers from subjective interpretation and lack of standardized analysis tools (Singh & Singh, 2020).

Existing AI tools for thermography are often:

- Research prototypes without reusable, documented code.
- Based solely on fine-tuning CNNs — prone to overfitting on small datasets (Roslidar et al., 2019).
- Lacking modularity, making extension or comparison difficult.

ThermoBreastAI addresses these gaps by providing:

1. A **reusable, pip-installable Python library** — not a monolithic application.
2. A **hybrid AI architecture** optimized for small datasets: frozen CNNs + interpretable ML.

3. **Built-in evaluation metrics** aligned with clinical needs (e.g., sensitivity prioritized over accuracy).
4. **Full test coverage and documentation** — enabling reproducibility and community contribution.

State of the Field

The application of artificial intelligence to breast thermography has gained traction in recent years, with studies demonstrating promising results using both classical machine learning and deep learning approaches.

Early work by (Vasconcelos et al., 2018) used statistical interval features and classical classifiers (e.g., SMO, Random Forest) on the HC/UFPE dataset, achieving 93.4% accuracy in binary classification. More recent studies have employed end-to-end CNNs: (Ekici & Jawzal, 2020) used a custom CNN architecture on UFF data to achieve 95.8% accuracy, while (Roslidar et al., 2019) and (Chaves et al., 2020) applied fine-tuning to models like DenseNet and VGG19, reporting accuracies up to 100% and 77.5%, respectively.

However, these approaches often suffer from critical limitations:

- **Lack of reproducibility**: Code is rarely open-sourced or packaged for reuse.
- **Overfitting risk**: Fine-tuning on small medical datasets (<1000 images) can lead to inflated performance metrics.
- **Monolithic design**: Most implementations are scripts or notebooks, not modular libraries.

ThermoBreastAI advances the field by:

- Releasing a **fully open-source, tested, and documented library**.
- Introducing a **hybrid architecture** that freezes CNN weights to prevent overfitting — a design choice validated by 100% sensitivity on clinical data.
- Providing **standardized evaluation metrics** (accuracy, sensitivity, specificity) and **automated hyperparameter tuning** via `GridSearchCV`.

Software Description / Functionality

ThermoBreastAI offers a comprehensive suite of tools for preprocessing, feature extraction, classification, and evaluation of breast thermograms. It combines an intuitive Python interface with robust machine learning backends.

Key Features

- **Preprocessing**: Automatic stripping of thermal metadata and resizing to 224x224 pixels for compatibility with pre-trained CNNs.
- **Feature Extraction**: Uses frozen CNN backbones (pre-trained on ImageNet) to extract robust, generalizable features — mitigating overfitting on small medical datasets.
- **Classification**: Supports Random Forest, KNN, and Naive Bayes with automated hyperparameter tuning via `GridSearchCV`.
- **Evaluation**: Computes standard clinical metrics: accuracy, sensitivity, specificity, and confusion matrix — prioritizing sensitivity for screening applications.
- **Extensibility**: Designed for easy integration of new CNN architectures or classifiers via subclassing.

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- **Reproducibility**: Includes sample data and unit tests (pytest) to validate installation and core functionality.

Example Usage

```python
from thermobreastai import HybridClassifier


# Initialize model
model = HybridClassifier(cnn_model='ResNet', classifier='RandomForest')


# Train on your dataset
image_paths = ["path/to/malignant.jpg", "path/to/healthy.jpg"]
labels = [1, 0]  #1 = malignant, 0 = healthy
model.fit(image_paths, labels)


# Predict and evaluate
test_paths = ["path/to/new_image.jpg"]
prediction = model.predict(test_paths)
print(f"Diagnosis: {'Malignant' if prediction\[0] == 1 else 'Healthy'}")


# Optional: Evaluate performance
# metrics = model.score(test_paths, test_labels)
# print(f"Sensitivity: {metrics['sensitivity']:.3f}")
```

Installation & Testing

```python
pip install git+https://github.com/AdriellySayonara/testedeploycanceria


# Run tests
pip install pytest
python -m pytest tests -v
```

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